ICT133

Proctored Online Exam - January Semester 2025

Structured Programming

Thursday, 17 April 2025 **01:00 pm - 03:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **6** questions.
2. Answer all parts of each question in the corresponding answer box.
3. All answers must be typed in the answer box and submitted in Examena.
4. This is a/an **Closed-book** examination.
5. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.
6. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.
7. **One** A4-sized (double-sided) hardcopy Cheat Sheet allowed.

At the end of the examination.
To end the examination, please click on the "Submit and end exam" button to submit your answers.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A company wants to categorize its employees based on their years of experience.

- Employees with less than 2 years of experience are considered "Junior".
- Employees with 2 or more years of experience, but less than 5 years, are considered "Mid-level".
- Employees with 5 or more years of experience are considered "Senior".

Write a Python program that takes an employee's years of experience as input and uses selection statements to determine and print the employee's category ("Junior", "Mid-level", or "Senior"). Assume the user will enter positive numbers for years of experience.

Sample run 1:

Enter years of experience: **3.5**

Employee category is Mid-level

Sample run 2:

Enter years of experience: **5**

Employee category is Senior

(8 marks)

Question 2

Write a Python program that uses **for** loops to print the input string “12345” in the following pattern.

```
1
12
123
1234
12345
```

Assuming user enter “hello” as input, your program output will be:

```
h
he
hel
hell
hello
```

(12 marks)

Question 3

Question 3 has 2 parts: Q3a & Q3b. Answer all 2 parts of this question.

Question 3a

Write a function deductWords(word1: str, word2: str) -> list that perform the following:

- Employs a process to creates a list of remaining characters in word1 after removing ALL occurrences of characters from word2 (case-sensitive).
- Parameters/arguments:
    - word1: The first word (from which characters are removed).
    - word2: The second word (characters to remove).
- Hence, the function returns a list containing the remaining characters of word1. If all characters of word1 are removed by word2, then return an empty list.

<u>Sample runs:</u>

```
print( deductWords("apple", "ap") )          # Output: ['l', 'e']
print( deductWords("banana", "ban") )        # Output: [ ]
print( deductWords("banana", "Bananas") )    # Output: ['b']
print( deductWords("hello", "world") )       # Output: ['h', 'e']
```

(15 marks)

Question 3b

Making use of the function in Q3(a), write a Python program that will prompt the user to enter 2 words, and apply the logic to display the remaining characters of word1, if any, otherwise, display “Nothing left”.

Your program should repeatedly ask the user for words, until the user enters “X” or “x” as the 1st word.

<u>Sample program runs (for reference only):</u>

Enter 1st word: **apple**
Enter 2nd word: **ap**
2 remaining character(s): l, e

Enter 1st word: **banana**
Enter 2nd word: **Bananas**
1 remaining character(s): b

Enter 1st word: **banana**
Enter 2nd word: **ban**
Nothing left

Enter 1st word: **X**
bye

(15 marks)

Question 4

Apply the logic to solve and produce the output of the 5 calls (line 12 to 16) to the guessWhat( ) function.

```
1  def guessWhat(numList1, numList2):
2      result = [ ]
3      for i in range(len(numList1)):
4          if i < len(numList2):
5              result.append(numList1[i] * numList2[i])
6          else:
7              result.append(0)
8
9      return result
10
11
12 print( guessWhat ( [1,2], [3,4] ) )
13 print( guessWhat ( [2,3,4], [3,4,5] ) )
14 print( guessWhat ( [4,4,4], [3,3] ) )
15 print( guessWhat ( [2,3], [1,2,3] ) )
16 print( guessWhat ( [1], [] ) )
```

(10 marks)

Question 5

Trace the given program below and **show the content of newDict (line 12) for each iteration of the loop.**

```
1  wordList = ['apple','banana','Apple','guava','cherry',
2              'apricot','Blackberry','blueberry']
3
4  newDict = { }
5  for word in wordList:
6      key = word[0].lower()
7      if key in newDict:
8          newDict[ key ] += 1
9      else:
10         newDict[ key ] = 1
11
12     print( newDict )
```

(10 marks)

Question 6

In ICT133, to promote active learning and facilitate in-class discussion and case study, students are required to read through Study Units and attempt 3 Pre-Class Quizzes

(PCQ).

For each PCQ, 3 multiple attempts are allowed up to the closing date and the highest marks will be the final score.

Student information is stored in a text file named “student.txt”. Each line contains the PI number. Below is an example of a “student.txt” file:

```
Z123456
G387230
S990234
W206574
```

To obtain the PCQ score, refer to the text file “assessment.txt”. Each line of the file contains information in this order: <PI number>,<PCQ number>,<score> with commas as separator. Assume all PI numbers, PCQ numbers and scores are valid.

Below is an example of a “assessment.txt” file:

```
Z123456,PCQ1,85
W206574,PCQ1,50
W206574,PCQ1,80
W206574,PCQ2,92.5
S990234,PCQ1,95
Z123456,PCQ1,100
W206574,PCQ2,90
S990234,PCQ2,95
Z123456,PCQ2,92.5
S990234,PCQ3,80
S990234,PCQ3,95
S990234,PCQ3,87.5
S990234,PCQ3,100
Z123456,PCQ3,87.5
```

By employing structured programming principle, develop Python functions, such that the following Python applications can execute.

```python
def main():
    result = readStudentData()
    loadAssessmentData(result)
    writeSummary(result)

```

Question 6a

Develop the function *readStudentData ( ): -> dict*

- This function reads the file “student.txt” and solves the task of collating the information into a dictionary and returns the dictionary.
- The key to the dictionary is the PI number, and the value is defaulted to:

  { 'PCQ1' : [ ], 'PCQ2': [ ], 'PCQ3': [ ] }

- Hence, the dictionary “result” in line 2 of main() should look like:

```
{
'Z123456': {'PCQ1': [], 'PCQ2': [], 'PCQ3': []},
'G387230': {'PCQ1': [], 'PCQ2': [], 'PCQ3': []},
'S990234': {'PCQ1': [], 'PCQ2': [], 'PCQ3': []},
'W206574': {'PCQ1': [], 'PCQ2': [], 'PCQ3': []}
}
```

(6 marks)

Question 6b

Write the function *loadAssessmentData (result: dict)*

- This function updates the given collection “result” with the student’s PCQ score read from the file “assessment.txt”.
- All PCQ’s scores will be recorded into its list. If there are already 3 scores for that PCQ, the 4th attempt will not be recorded. For example, student S990234, his score of 100 for PCQ3 is to be discarded.
- The dictionary result should be updated the following after line 3 of main():

```
{
'Z123456': {'PCQ1': [85.0, 100.0], 'PCQ2': [92.5], 'PCQ3': [87.5]},
'G387230': {'PCQ1': [], 'PCQ2': [], 'PCQ3': []},
'S990234': {'PCQ1': [95.0], 'PCQ2': [95.0], 'PCQ3': [80.0, 95.0, 87.5]},
'W206574': {'PCQ1': [50.0, 80.0], 'PCQ2': [92.5, 90.0], 'PCQ3': []}
}
```

(12 marks)

Question 6c

Write the function *writeSummary (result: dict)*

- Given the collection “result”, this function writes the overall result to “summary.txt”
- Each line in “summary.txt” will contain the PI number, PCQ number, and the final score, separated by space
- Sample output file “summary.txt”, after line 4 of main():

Z123456 PCQ1 100.0
Z123456 PCQ2 92.5
Z123456 PCQ3 87.5
G387230 PCQ1 No attempt
G387230 PCQ2 No attempt
G387230 PCQ3 No attempt
S990234 PCQ1 95.0
S990234 PCQ2 95.0
S990234 PCQ3 95.0
W206574 PCQ1 80.0
W206574 PCQ2 92.5
W206574 PCQ3 No attempt

- If student has attempted a PCQ, the final score is the highest score among the attempts.
- If student has not attempted a PCQ, print the final score as "No attempt".

(12 marks)

----- END OF PAPER -----